

# Qtum: promoting an efficient blockchain economy with Google Cloud

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## ABOUT QTUM

By running on Google Cloud, Qtum delivers an open source blockchain platform while reducing cloud costs, lowering update and publication costs, and providing an efficient and secure environment that allows customers to move from bitcoin to a more efficient blockchain.



Q T U M

## About Qtum

Qtum is a decentralized, open source public blockchain project that aims to combine the advantages of the Bitcoin and Ethereum digital currencies and solve the shortcomings of existing blockchain systems.

Industries: Technology

Location: Singapore

## Google Cloud results

- Reduced cloud costs by 30% relative to a previous provider
- Optimized use of Google Cloud to reduce update publication times from two weeks to four days
- Lowered data transfer bill by 18%

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Power source that aims to extend the blockchain into finance, social networking, gaming and the Internet of Things.

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Compute Engine

(<https://cloud.google.com/compute/>)

BigQuery (<https://cloud.google.com/bigquery/>)

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Blockchain is high on the list of technologies transforming business today. PwC noted in a [2018 Global Blockchain survey](https://www.pwc.com/gx/en/industries/technology/blockchain/blockchain-in-business.html).

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of 600 executives in 15 territories that, “as a distributed, tamperproof ledger, a well-designed

blockchain doesn't just cut out intermediaries, reduce costs, and increase speed and reach. It also offers greater transparency and traceability for many business processes."

The researchers note 84% of respondents' organizations had some involvement with blockchain, while Gartner forecasts (<https://www.gartner.com/en/webinars/3878710/blockchain-potential-and-pitfalls#:~:text=Gartner%20now%20forecasts%20that%20blockchain,in%20the%20last%20two%20years.>) the technology to generate US\$3 trillion in annual business value by 2030.

Qtum (<https://qtum.org/en>) [pronounced Quantum] is a growing presence in the blockchain market. Qtum is a distributed, open source blockchain platform. Its aims include combining the best features of leading digital currencies Ethereum and Bitcoin to provide a secure cryptocurrency that can be easily "mined," or earned without financial investment.

Two organizations are primarily responsible for the development and promotion of the Qtum blockchain: the Qtum Blockchain Foundation, a not-for-profit based in Singapore with a mission of developing and enhancing the codebase and promoting the product's use; and the Qtum Chain Foundation (<https://qtum.org>), also based in Singapore, that leads development, improves the Qtum Core blockchain, and organizes public relations, marketing, and business development to grow the platform's popularity.

In a whitepaper, Qtum's founders say they "united community development forces, dug into the underlying technology and developed and implemented a series of technologies through the 'value transfer protocol' upgrade and other innovative solutions to build a sustainable public blockchain."

The founders ultimately aim to integrate the blockchain into industries such as finance, social networking, gaming, and the Internet of Things.



## **Cloud a no-brainer**

The highly distributed, compute- and data-intensive nature of the Qtum platform made running in the cloud a no-brainer. Qtum established its cost, performance, and resource requirements and began running on a cloud operated by an organization headquartered in China, then on a cloud run by an organization based in the United States. However, to meet its stability, cost, and data processing and

analysis requirements, the Qtum Foundation turned to Google Cloud.

“We already had some seed nodes [nodes that incorporate the latest information about the blockchain and provide a connection point for new nodes] running in Google Cloud when we ran most of our workloads in other clouds,” says Miguel Palencia, Chief Information Officer, Qtum. “Our experience with these nodes encouraged us to explore Google Cloud further,” he says.

“Analyzing these transactions and this data wasn’t possible with anything but BigQuery, so the service is a game changer for us.”

—Miguel Palencia, Chief  
Information Officer, Qtum

**Dedicated fiber network key to  
decision-making**

Qtum's review revealed Google Cloud (<https://cloud.google.com/>) provided the flexibility and stability the platform needed, and its leadership and technology teams were particularly impressed by the dedicated connections between key markets provided by Google's global fiber network.

Qtum is now running the platform—including all seed and primary nodes—in a Google Cloud infrastructure distributed across Hong Kong, the United States, and Europe. The organization uses Compute Engine (/compute) to host a template pre-installed with services used to create new nodes, enhancing the efficiency of a process that supports a marketplace offering—an environment that enables developers to start building smart contracts—first launched in 2018. This offering is an environment for developers to start building smart contracts.

Once Qtum launches a new node and synchronizes it with its peer-to-peer network, it has the base platform for anything it needs.

The organization also uses a BigQuery (/bigquery) data warehouse to enable analysis of data generated within the Bitcoin blockchain. Each “block”—created every 10 minutes—incorporates a large volume of transactions and data. “Analyzing these transactions and this data wasn't possible with anything but BigQuery, so the service is a game changer for us,” says Palencia.

Qtum is using the templates created within Google Cloud to build a range of products and services. One

product allows users to search the blockchain, while another enables them to track the movement of Bitcoin on and off the exchanges that allow customers to trade cryptocurrencies or digital currencies for other assets.

The latter service sorts and classifies hundreds of thousands of cryptocurrency investor addresses into different categories and identifies the addresses used by exchanges. Tracking, analyzing, and reporting on the movement of Bitcoin into exchanges allows Qtum to rank exchanges on a weekly or monthly basis. This enables the cryptocurrency community to better understand blockchain operations and how markets behave.

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*—Miguel Palencia, Chief  
Information Officer, Qtum*

## **Tracking large transactions**

BigQuery-powered analysis also enables Qtum to track transactions of 50 Bitcoin or more, allowing users to track the movements of investors with large accounts and make their own decisions accordingly. The business makes live data about large bitcoin transfers to and from exchanges publicly available at <https://chain.info> (<https://chain.info>).

With Google Cloud, Qtum has grown while reducing its cloud spend. “As a startup, we’re looking to provide a good experience and good service while being efficient with our expenses,” says Palencia. “Google Cloud is allowing us to do that.” The organization has reduced its costs by 30% compared to a previous cloud provider and lowered its data transfer bill by 18% relative to another provider—all while leaving behind the stability issues that compromised its operations on other services.

Qtum has improved its processes and knowledge of Google Cloud, enabling it to reduce the time required to publish feature, security, and mandatory updates to the market from two weeks to four days.

“Overall, Google Cloud saves us effort, headaches and money—and we haven’t seen any downtime on our main website for a long time. We believe Google Cloud will continue to play a major role on our platform and everything we are going to be running in the future.”

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**Saving “effort, headaches, and money”**

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The business is now looking at Kubernetes to build clusters to improve the performance and reliability of some services and enhance scalability. “Scaling Qtum nodes with Kubernetes can create ‘supernodes’ faster and more resilient than regular nodes,” says Palencia. “This will increase performance and in turn create a stronger back end for future decentralized applications running on Qtum.”